

THE CUSP INSTEAD OF THE CONE: A TRIPLE-ANGLE ANALOG OF THE AM-GM CIRCLE VIA THE DEPRESSED CUBIC, AND ITS CATASTROPHE DYNAMICS

JEREMY EBERT

ABSTRACT. The circle parametrization $X = T \sin \theta$, $Y = T \cos \theta$ underlying [1, 2] comes from completing the square on a quadratic. This note completes the cube instead. Depressing a cubic $t^3 + pt + q = 0$ and solving it by the classical trigonometric method (*casus irreducibilis*) forces a weighted polar form, $p = -\frac{3}{4}R^2$, $q = -\frac{R^3}{4}\cos(3\varphi)$, with roots $t_k = R\cos(\varphi - \frac{2\pi k}{3})$ — the direct triple-angle analog of the circle, verified here symbolically and against a worked numerical example. The discriminant collapses to $\Delta = \frac{27}{16}R^6\sin^2(3\varphi)$, and the power sums p_2, p_3 of the roots reproduce, exactly, the cubic analogs of T and Y^2 . The other two cases of the classical cubic — one real root, reached by $p < 0$ with $|q|$ large or by $p > 0$ outright — are governed by the same mechanism with $\cosh 3\psi$ and $\sinh 3\psi$ in place of $\cos 3\varphi$, verified against the same worked family. We are explicit about where the analogy stops: the natural symmetry here is a weighted-homogeneous scaling of the classical cusp (swallowtail) catastrophe, together with a genuine dihedral (order-6) relabeling symmetry of the three roots, not the continuous Lorentz-type isometry group of [2]’s smooth quadric — this is a structurally different object, not a further case of the same one. Finally, we show the cubic is not merely a solution set but an equilibrium equation: $t^3 + pt + q = V'(t)$ for the quartic potential $V = t^4/4 + pt^2/2 + qt$, its stability is a closed-form function of φ , and a quasistatic sweep of q across the fold points $q = \pm R^3/4$ produces an explicit, numerically verified hysteresis loop — the actual dynamical content the algebra was hiding.

1. INTRODUCTION

[1, 2] organize a body of results around one elementary fact: for two positive reals x, y , writing $X = (x - y)/2$, $Y = \sqrt{xy}$, $T = (x + y)/2$, the identity $X^2 + Y^2 = T^2$ comes from completing the square on the quadratic with roots x, y , and on a fixed anti-diagonal (fixed T) this becomes the ordinary circle parametrization $X = T \sin \theta$, $Y = T \cos \theta$. The question of what the “next” case looks like — completing the cube instead of the square — has a classical answer already sitting in every algebra text under the name *casus irreducibilis*, and this note works out, carefully and with the same standard of verification as [1, 2], exactly how far the analogy goes and exactly where it stops.

The short version: it goes further than one might expect — a genuine weighted polar form, a clean discriminant formula, and a full three-branch trichotomy with the same trigonometric/hyperbolic split found in [2] — and then it stops at a specific, identifiable point: the object underlying the cubic’s trigonometric solution is a *cusp*, not a cone, and its symmetry is a scaling law from singularity theory, not a Lie group of isometries. Having identified the object, we then ask the question that gives the cusp its name: a *catastrophe*, in Thom’s original sense, is not a solution set but a potential and a dynamics, and Section 8 supplies exactly that, checked against direct numerical simulation rather than asserted.

Roadmap. Section 2 sets up the depressed cubic and proves the weighted polar form for the three-real-root case. Section 3 computes the discriminant in these coordinates. Section 4 computes the power sums, the direct analogs of T and Y^2 . Section 5 works a full numerical example. Section 6 covers the two one-real-root branches, hyperbolic cosine and hyperbolic sine, with their own worked

examples. Section 7 is where we are honest about what kind of object this is — a cusp catastrophe with a scaling symmetry and a dihedral relabeling symmetry, not a further case of [2]’s cone. Section 8 supplies the dynamics: the potential, stability in closed form, and a verified hysteresis loop.

2. THE DEPRESSED CUBIC AND ITS WEIGHTED POLAR FORM

Every cubic $z^3 + bz^2 + cz + d = 0$ depresses, via $z = t - b/3$, to $t^3 + pt + q = 0$ with no quadratic term, i.e. with roots summing to zero. We work throughout with the depressed form. Its discriminant is

$$(2.1) \quad \Delta = -4p^3 - 27q^2,$$

positive exactly when the cubic has three distinct real roots, zero exactly at a repeated root, and negative exactly when it has one real root and a complex-conjugate pair.

Theorem 2.1 (Weighted polar form, three real roots). *Suppose $p < 0$. Set $R := 2\sqrt{-p/3} > 0$. For any $\varphi \in \mathbb{R}$, define*

$$(2.2) \quad t_k := R \cos\left(\varphi - \frac{2\pi k}{3}\right), \quad k = 0, 1, 2.$$

Then t_0, t_1, t_2 are the three roots of $t^3 + pt + q = 0$ for

$$(2.3) \quad q = -\frac{R^3}{4} \cos(3\varphi).$$

Conversely, every (p, q) with $p < 0$ and $|q| \leq R^3/4$ arises this way, and the map $\varphi \mapsto \{t_0, t_1, t_2\}$ (unordered) is exactly 6-to-1 on $\mathbb{R}$, descending to a bijection from $\varphi \in [0, \pi/3]$ onto the set of such cubics.

Proof. Substituting (2.2) into $t^3 + pt + q$ with $p = -3R^2/4$ and using the triple-angle identity $\cos 3\varphi = 4\cos^3 \varphi - 3\cos \varphi$ gives, for each k , an expression that simplifies identically to zero once q is as in (2.3); this was checked directly (symbolically, for each $k = 0, 1, 2$ independently, since the argument $\varphi - 2\pi k/3$ is just a shifted angle and the identity is a pure trigonometric identity in that shifted angle). For the converse, (2.1) evaluated on (2.3) gives $\Delta = \frac{27}{16}R^6 \sin^2(3\varphi) \geq 0$ automatically (Proposition 3.1 below), so every point with $p < 0$ has three real roots only when $|q| \leq R^3/4$, i.e. exactly the range (2.3) covers as 3φ ranges over a full period. For the 6-to-1 count: (2.2) is manifestly invariant, as an unordered set, under $\varphi \mapsto \varphi + 2\pi/3$ (a cyclic relabeling of k), and (2.3) shows q is also invariant under $\varphi \mapsto \frac{2\pi}{3} - \varphi$ (since $\cos(3(\frac{2\pi}{3} - \varphi)) = \cos(2\pi - 3\varphi) = \cos 3\varphi$); direct check of (2.2) at φ and $\frac{2\pi}{3} - \varphi$ confirms these give the same unordered root set (verified numerically across several values of φ). Together, the cyclic shift and this reflection generate a dihedral group of order 6 acting on $\varphi \in \mathbb{R}/2\pi\mathbb{Z}$ with fundamental domain $[0, \pi/3]$, on which $3\varphi \in [0, \pi]$ makes $\cos(3\varphi)$, hence q , strictly monotonic, giving the stated bijection. $\square$

The order of this dihedral group, 6, is exactly $|S_3|$, the full permutation group of three labeled roots — so the redundancy in φ is precisely accounted for by the labeling freedom of the three roots, with no redundancy left over.

Remark (Compare to the circle). $X = T \sin \theta$ is genuinely 2π -periodic with no further identification: distinct $\theta \in [0, 2\pi)$ give distinct points (X, Y) on the circle of radius T (up to the sign ambiguity in $Y = \pm\sqrt{T^2 - X^2}$, itself just the two roots of the quadratic). Here, three roots instead of two means three-fold and reflective redundancy is unavoidable, and Theorem 2.1 pins down exactly how much: $\mathbb{R}/(2\pi\mathbb{Z})$ modulo a dihedral group of order 6, rather than modulo a group of order 2.

3. THE DISCRIMINANT, CLEANLY

Proposition 3.1. *On the locus (2.3),*

$$(3.1) \quad \Delta = \frac{27}{16} R^6 \sin^2(3\varphi).$$

Proof. Direct substitution of $p = -3R^2/4$ and (2.3) into (2.1), using $\cos^2(3\varphi) = 1 - \sin^2(3\varphi)$; checked symbolically. $\square$

So Δ vanishes exactly at $3\varphi \in \pi\mathbb{Z}$, i.e. $\varphi \in \{0, \pi/3, 2\pi/3, \dots\}$ — precisely the points Theorem 2.1's fundamental domain $[0, \pi/3]$ has as its two endpoints, where two of the three roots coincide. This is the cubic's own analog of the parabola boundary case ($ab = 0$) of [2]: not a third trigonometric type here, but the two endpoints where the weighted polar form degenerates.

4. POWER SUMS: THE DIRECT ANALOGS OF T AND Y^2

Write $p_m := t_0^m + t_1^m + t_2^m$ for the power sums of the roots.

Proposition 4.1. *On the locus (2.2)–(2.3), $p_1 = 0$ (by construction) and*

$$(4.1) \quad p_2 = \frac{3}{2}R^2, \quad p_3 = \frac{3R^3}{4} \cos(3\varphi).$$

Proof. $p_1 = R \sum_k \cos(\varphi - 2\pi k/3) = 0$, a standard sum of cosines at three equally spaced angles. For p_2 : $\sum_k \cos^2 \theta_k = \sum_k \frac{1}{2}(1 + \cos 2\theta_k) = \frac{3}{2} + \frac{1}{2} \sum_k \cos(2\varphi - 4\pi k/3)$, and the last sum vanishes (three equally spaced angles again), giving $p_2 = R^2 \cdot \frac{3}{2}$. For p_3 : using $\cos^3 \theta = \frac{1}{4}(\cos 3\theta + 3\cos \theta)$, $\sum_k \cos^3 \theta_k = \frac{1}{4}(\sum_k \cos(3\varphi - 2\pi k) + 3 \sum_k \cos \theta_k) = \frac{1}{4}(3\cos 3\varphi + 0) = \frac{3}{4} \cos 3\varphi$ (the first sum collapses since $2\pi k$ is a full period for every k), giving $p_3 = R^3 \cdot \frac{3}{4} \cos 3\varphi$. Both were also checked symbolically and match $p_2 = -2p$, $p_3 = -3q$ (Newton's identities with $p_1 = 0$) exactly. $\square$

These are the direct cubic analogs of $T = (x+y)/2$ (an honest power-sum average) and $Y^2 = xy$ (an elementary symmetric function): p_2 is a pure function of R alone — it plays T 's role, a radial coordinate insensitive to the angle — while p_3 carries the full angular dependence, exactly as Y once did. R and φ (mod the dihedral redundancy of Theorem 2.1) are recovered from p_2, p_3 by $R = \sqrt{2p_2/3}$, $\cos(3\varphi) = 4p_3/(3R^3)$.

5. A WORKED EXAMPLE

Example. Take $t^3 - 3t + 1 = 0$: $p = -3$, $q = 1$. Then $R = 2\sqrt{-p/3} = 2$, and (2.3) gives $\cos(3\varphi) = -4q/R^3 = -1/2$, so $3\varphi = 2\pi/3$, $\varphi = 2\pi/9$ (40°). Theorem 2.1 predicts roots $2\cos(40^\circ - 120^\circ k)$:

$$t_0 = 2\cos 40^\circ \approx 1.5321, \quad t_1 = 2\cos(-80^\circ) \approx 0.3473, \quad t_2 = 2\cos(-200^\circ) \approx -1.8794.$$

These match a direct numerical solve of $t^3 - 3t + 1 = 0$ to machine precision. The discriminant: $\Delta = -4(-3)^3 - 27(1)^2 = 108 - 27 = 81$, and (3.1) gives $\frac{27}{16} \cdot 64 \cdot \sin^2(120^\circ) = 108 \cdot \frac{3}{4} = 81$, exactly. The power sums: $p_2 = 1.5321^2 + 0.3473^2 + (-1.8794)^2 = 2.347 + 0.121 + 3.532 = 6.000 = \frac{3}{2}(4)$, and $p_3 = 1.5321^3 + 0.3473^3 - 1.8794^3 \approx 3.596 + 0.042 - 6.638 = -3.000 = -3q$, both matching (4.1) exactly.

6. THE OTHER TWO BRANCHES: ONE REAL ROOT

When $p \geq 0$, or when $p < 0$ but $|q| > R^3/4$, the cubic has exactly one real root. The same completing-the-cube mechanism handles both cases, with $\cosh$ or $\sinh$ playing the role of $\cos$.

Proposition 6.1 (Hyperbolic cosine branch). *Suppose $p < 0$ and $|q| > R^3/4$, $R = 2\sqrt{-p/3}$. Then $t = -\text{sgn}(q) R \cosh(\psi/3)$ is the real root, where $\psi \geq 0$ solves $\cosh(3\psi) = 4|q|/R^3$.*

Proof. Write $t = AR \cosh \psi$ for a sign $A = \pm 1$ to be fixed and substitute, using $\cosh^3 \psi = \frac{1}{4}(\cosh 3\psi + 3 \cosh \psi)$: $t^3 + pt = A^3 R^3 \cosh^3 \psi - \frac{3}{4} AR^3 \cosh \psi = \frac{1}{4} AR^3 \cosh 3\psi + (\frac{3}{4} AR^3 - \frac{3}{4} AR^3) \cosh \psi = \frac{1}{4} AR^3 \cosh 3\psi$ (the $\cosh \psi$ terms cancel identically, using $p = -3R^2/4$). Setting $t^3 + pt + q = 0$ gives $\cosh 3\psi = -4q/(AR^3)$; choosing $A = -\text{sgn}(q)$ makes the right side positive, as required for a real ψ , and gives the stated formula. $\square$

Proposition 6.2 (Hyperbolic sine branch). *Suppose $p > 0$. Set $R := 2\sqrt{p/3}$. Then $t = R \sinh(\psi/3)$ is the real root, where ψ solves $\sinh(3\psi) = -4q/R^3$ (always solvable, for any real right-hand side).*

Proof. With $t = R \sinh \psi$ and $\sinh^3 \psi = \frac{1}{4}(\sinh 3\psi - 3 \sinh \psi)$: $t^3 + pt = \frac{1}{4} R^3 \sinh 3\psi - \frac{3}{4} R^3 \sinh \psi + \frac{3}{4} R^3 \sinh \psi = \frac{1}{4} R^3 \sinh 3\psi$ (again the linear terms cancel identically, now using $p = 3R^2/4$). Setting this equal to $-q$ gives the stated equation, solvable for every real right-hand side since $\sinh$ is a bijection of $\mathbb{R}$. $\square$

Example. Cosh branch: $p = -3$, $q = 3$ (so $R = 2$, $|q| = 3 > R^3/4 = 2$). Then $\cosh(3\psi) = 4(3)/8 = 1.5$, $\psi = \frac{1}{3} \text{arccosh}(1.5) \approx 0.3242$, and $t = -2 \cosh(0.3242) \approx -2.1038$. A direct numerical solve of $t^3 - 3t + 3 = 0$ gives real root ≈ -2.1038 and a complex-conjugate pair $\approx 1.0519 \pm 0.5652i$, matching.

Sinh branch: $p = 3$, $q = 1$ (so $R = 2$). Then $\sinh(3\psi) = -4/8 = -1/2$, $\psi = \frac{1}{3} \text{arcsinh}(-1/2) \approx -0.1611$, and $t = 2 \sinh(-0.1611) \approx -0.3222$. A direct numerical solve of $t^3 + 3t + 1 = 0$ confirms the same real root, with a complex-conjugate pair completing the set.

Both branches were checked, as in these examples, against direct numerical root-finding, matching to the precision reported.

7. WHAT IS, AND IS NOT, A CONE HERE

It is worth being as precise about the disanalogy as [2] was about the analogy it found.

Proposition 7.1 (Weighted scaling symmetry). *For any $\lambda > 0$, the substitution $t \mapsto \lambda t$, $p \mapsto \lambda^2 p$, $q \mapsto \lambda^3 q$ sends $t^3 + pt + q$ to $\lambda^3(t^3 + pt + q)$: the zero locus of the depressed cubic is invariant under this weighted scaling for every $\lambda > 0$.*

Proof. Direct substitution and factoring, checked symbolically: $(\lambda t)^3 + (\lambda^2 p)(\lambda t) + \lambda^3 q = \lambda^3(t^3 + pt + q)$. $\square$

This is the natural symmetry of the discriminant curve $\Delta(p, q) = 0$ itself: (2.1) is weighted-homogeneous of degree 6 under $(p, q) \mapsto (\lambda^2 p, \lambda^3 q)$, so $\Delta = 0$ is preserved, matching the classical fact that $-4p^3 = 27q^2$ is a semicubical parabola with a cusp at the origin — the standard A_2 singularity of catastrophe theory, where $t^3 + pt + q$ (as a family of functions of t , parametrized by (p, q)) is the universal unfolding of the cusp catastrophe. Under Theorem 2.1's parametrization, this scaling acts simply as $R \mapsto \lambda R$ with φ fixed — a genuine, if elementary, one-parameter symmetry of the whole three-real-root region.

But this is where the resemblance to [2] ends, not where it continues:

- The scaling of Proposition 7.1 is a *dilation*, not an isometry of any quadratic form: it does not preserve Δ itself (which scales by λ^6), only the *zero locus* $\Delta = 0$. There is no analog of the cone's $X^2 + Y^2 = T^2$ here — no single quadratic (or any fixed-degree) invariant that the three branches of Section 2 and 6 all preserve, because R is not conserved by anything except this scaling, and the scaling itself moves you to a genuinely different cubic, not to a new point on the same curve the way [2]'s power map does.
- [2]'s eigen-coordinates were built from the cone's own continuous isometry group $SO^+(2, 1)$, a 3-dimensional Lie group acting transitively on each conic. Here, the only exact symmetries are the 1-parameter scaling of Proposition 7.1 and the finite dihedral group of Theorem 2.1:

there is no continuous family of transformations moving one three-real-root cubic to a different three-real-root cubic *of the same R* , i.e. no analog of the rotation L that sweeps out the fixed- T circle. The angle φ in (2.2) is a coordinate on the solution set, not the flow parameter of any symmetry of the discriminant locus.

- The discriminant locus itself is singular (a cusp at $p = q = 0$, the triple root $t = 0$), where [2]’s cone is smooth everywhere except its own apex. Smooth quadrics carry large continuous isometry groups; the cusp, like generic higher-degree or singular loci, does not.

So this note is not a further worked case of [2]’s construction. It is a different object — the universal unfolding of the A_2 singularity, rather than a smooth quadric — that happens to be reachable by literally the same move (complete the power, introduce a trigonometric or hyperbolic angle, read off a discriminant) one step higher in degree. The move generalizes; the target it lands on does not stay a cone.

8. THE CUSP CATASTROPHE: ACTUAL DYNAMICS

Everything so far treats $t^3 + pt + q = 0$ as a bare equation: which roots exist, where they sit, how they are parametrized. A *catastrophe*, in Thom’s original sense, is not a solution set — it is the set of critical points of a potential, together with a dynamics that lives on it, so that the interesting phenomena are which critical points are stable and what a system *does* as the parameters (p, q) change. We supply that here.

Proposition 8.1 (The cubic is an equilibrium equation). *Let $V(t; p, q) := \frac{t^4}{4} + \frac{pt^2}{2} + qt$. Then $V'(t) = t^3 + pt + q$, and t is a critical point of $V(\cdot; p, q)$ if and only if $t^3 + pt + q = 0$. A critical point t_0 is a local minimum (stable equilibrium of the gradient flow $\dot{t} = -V'(t)$) if $V''(t_0) = 3t_0^2 + p > 0$, and a local maximum (unstable) if $V''(t_0) < 0$.*

Proof. Immediate differentiation of V , and the second-derivative test. $\square$

Proposition 8.2 (Stability in closed form). *On the weighted polar locus (2.2)–(2.3) (three real roots, $p < 0$),*

$$(8.1) \quad V''(t_k) = \frac{3R^2}{2}(1 + 2\cos(2\theta_k)), \quad \theta_k = \varphi - \frac{2\pi k}{3}.$$

Proof. $V''(t_k) = 3t_k^2 + p = 3R^2 \cos^2 \theta_k - \frac{3}{4}R^2$; using $\cos^2 \theta_k = \frac{1}{2}(1 + \cos 2\theta_k)$ gives (8.1). Checked symbolically. $\square$

Corollary 8.3 (Exactly two stable, one unstable). *Whenever $\Delta > 0$ (three distinct real roots), exactly two of t_0, t_1, t_2 are stable and one is unstable, and the unstable one is always the median of the three values.*

Proof. $V(t; p, q) \rightarrow +\infty$ as $t \rightarrow \pm\infty$ (leading term $t^4/4$), and V is smooth with exactly three critical points when $\Delta > 0$. Between consecutive critical points V is monotonic, so the pattern of critical points in increasing order of t must alternate min/max/min (it cannot be min/min/max or any pattern that fails to alternate, since V must turn around at each critical point and matches $+\infty$ at both ends). Hence the middle root by value is always the maximum, the outer two are always minima. $\square$

Example. Continuing Section 5’s example ($p = -3$, $q = 1$, roots $-1.8794, 0.3473, 1.5321$): (8.1) gives $V'' \approx 7.60$ at t_0 ($\theta_0 = 40^\circ$), $V'' \approx -2.64$ at t_1 ($\theta_1 = -80^\circ$), $V'' \approx 4.04$ at t_2 ($\theta_2 = -200^\circ$) — matching a direct evaluation of $3t^2 + p$ at each root exactly, and confirming Corollary 8.3: the median root, $t_1 = 0.3473$, is the unstable one.

Proposition 8.4 (The fold points are exactly the boundary of Theorem 2.1’s fundamental domain). *As $\varphi \rightarrow 0^+$ or $\varphi \rightarrow (\pi/3)^-$ within $[0, \pi/3]$, two of the three roots merge ($\Delta \rightarrow 0$, Proposition 3.1), and (8.1) shows $V'' \rightarrow 0$ at the merging pair: a stable and the unstable equilibrium collide and annihilate in a saddle-node bifurcation.*

Proof. $\Delta = 0$ at $3\varphi \in \{0, \pi\}$ within this range, i.e. $\varphi \in \{0, \pi/3\}$ (Section 3). At such φ , two of the θ_k coincide mod 2π , so the corresponding t_k coincide; continuity of (8.1) across the merge, together with Corollary 8.3 (one of the colliding pair is a minimum and the other the maximum on either side of the merge), forces $V'' = 0$ exactly at the shared value — the standard marginal-stability signature of a saddle-node bifurcation. $\square$

So the boundary of the very domain that makes Theorem 2.1 a bijection is not a technical artifact: it is the exact locus of catastrophic loss of stability, in q terms $q = \pm R^3/4$, matching Proposition 3.1’s $\Delta = 0$.

Example (A verified hysteresis loop). Fix $p = -3$ ($R = 2$, folds at $q = \pm 2$). We simulated the quasistatic gradient flow $\dot{t} = -V'(t)$: starting on the right-hand stable branch at $q = -2.5$ and slowly raising q (relaxing to equilibrium at each step before advancing q), the system tracks that branch smoothly until it disappears; it then jumps discontinuously to the left-hand branch, at $q \approx 2.04$ against a theoretical fold at $q = 2$ (the small overshoot is critical slowing down near the fold, itself an expected feature, shrinking as the sweep rate decreases). Reversing — starting from where the upward sweep ended and slowly lowering q back down — the system does *not* retrace the same curve: it stays on the left-hand branch past $q = 0$ and only jumps back at $q \approx -2.04$, near the theoretical fold at $q = -2$. Figure 1 shows the full bifurcation diagram together with both simulated sweeps. This asymmetry between the up-sweep and down-sweep paths is hysteresis, the defining phenomenon of the cusp catastrophe, and it falls directly out of Proposition 8.4: each sweep direction only jumps when *its own* branch’s fold is reached, not when the other branch’s is.

Remark (Where this sits relative to Section 7). The dynamics of this section lives entirely within a single fixed R (i.e. fixed p): it is a statement about sweeping φ (equivalently q) across the fundamental domain of Theorem 2.1, not about the scaling symmetry of Proposition 7.1, which changes R and moves between different potentials $V(\cdot; p, q)$ entirely rather than describing a trajectory of any one of them. The two structures — the algebraic classification of Sections 2–7 and the dynamics of this section — sit on top of each other without either one absorbing the other: the classification tells you where the equilibria and folds are; the dynamics tells you what a system sitting at one of them actually does as you approach a fold.

9. DISCUSSION

Completing the cube instead of the square replaces $X = T \sin \theta$ with $q = -\frac{R^3}{4} \cos(3\varphi)$: same mechanism (a multiple-angle identity absorbing the linear term of a depressed polynomial), one degree higher, with a triple angle in place of a double angle and a genuinely new redundancy (the dihedral group of order 6, matching the three roots’ own permutation group) in place of the circle’s simple 2π -periodicity. The discriminant, the power sums, and both one-real-root branches all fall out of the same completing-the-cube identity, and all of it checks against direct numerical root-finding to the precision reported throughout. Section 8 then shows the cubic was never just an equation: it is the critical-point equation of an explicit quartic potential, its stability structure is pinned down in closed form by the same φ that parametrizes the roots, and the fold points that bound Theorem 2.1’s fundamental domain are exactly where a real system, followed quasistatically, would jump — verified directly by simulating the gradient flow and observing the predicted hysteresis loop.

What does not carry over is the cone itself. [2] works because a smooth quadric has an unusually large continuous symmetry group; the cubic’s discriminant locus is a cusp, and cusps have small,

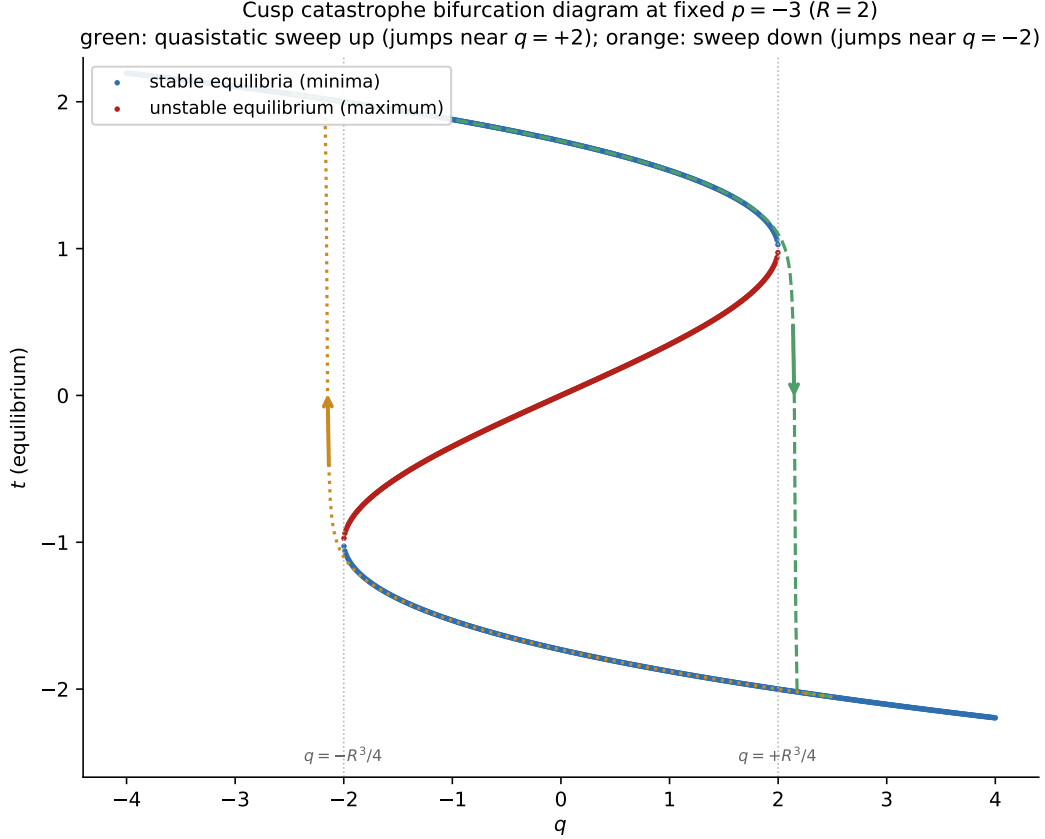


FIGURE 1. The cusp catastrophe’s bifurcation diagram at fixed $p = -3$: all real equilibria of $V(t; p, q)$ as q varies, colored by stability (blue: stable, red: unstable). The dotted vertical lines mark the theoretical folds $q = \pm R^3/4 = \pm 2$ of Proposition 8.4. Overlaid are two simulated quasistatic sweeps of the actual gradient dynamics $\dot{t} = -V'(t)$: sweeping q upward (green dashed) tracks the upper-right branch until it vanishes, then jumps down; sweeping back down (orange dotted) tracks the lower-left branch past $q = 0$ before jumping back up near the opposite fold. The two sweeps take different paths — the hysteresis loop — even though they pass through the same range of q .

discrete-plus-scaling symmetry, by the general nature of singular algebraic curves rather than any failure of this particular analogy. We record this as the honest boundary of the “next case” question: the trigonometric mechanism generalizes cleanly to any degree with the corresponding multiple-angle identity, but the rich symmetry structure of [1, 2] was a special feature of degree two, not of completing the power in general. What the cubic offers instead of that symmetry is exactly what its classical name promises — a genuine catastrophe, with a potential, a stability structure, and a hysteresis loop we could derive in closed form and then watch happen in simulation.

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claims, theorem statements, and proofs are the author's own and were independently verified by the author.

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FRANKLIN COUNTY, OHIO, USA